

CMP-7037A: Network Security and IoT
Assessment Two (CMP-7037A-25-SEM1-A 002)

A Hybrid Framework Utilizing Blockchain Integrity and AI-Driven Behavioral Analysis for E-Voting Simulation

Group 3

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Academic Year: 2025/2026

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1 Introduction

The integrity of the electoral process serves as the foundational pillar upon which modern democratic societies are built. As the global landscape undergoes a rapid "digital-first" transformation, the mechanisms of voting the literal voice of the people face unprecedented technological scrutiny. Traditional paper-based systems, while physically auditable and historically trusted, are increasingly viewed through the lens of logistical inefficiency. They are prone to localized human error, susceptible to physical tampering during the "chain of custody," and inherently slow in the age of instantaneous communication.

However, the first generation of electronic voting (e-voting) systems has frequently compromised fundamental security principles in favor of user convenience. Centralized infrastructures, where a single government entity or third-party vendor controls the database, have introduced catastrophic "single points of failure." Such architectures create a "honeypot" for state-level adversaries and cyber-criminals, where a single administrative breach can alter a national election's outcome without leaving a verifiable audit trail.

This project identifies a critical "trust gap" in existing digital voting frameworks and proposes a solution to the "E-voting Trilemma": the simultaneous achievement of security, privacy, and scalability. The challenge is not merely technical but philosophical; a system must be transparent enough for public verification yet private enough to ensure the sanctity of the secret ballot. This research posits that a single-layered defense is no longer sufficient.

By integrating Distributed Ledger Technology (DLT) specifically blockchain this project addresses the issue of data persistence and immutability. Once a vote is cast, it is mathematically sealed within a hash-chain. Yet, recognizing that an immutable ledger cannot distinguish between a legitimate human voter and an automated botnet, the project introduces a secondary layer of Artificial Intelligence (AI). This hybrid framework aims to protect both the storage (the ledger) and the entry (the behavior) of the electoral system, providing a resilient proof-of-concept for future digital governance.

2 Aims and Objectives

2.1 Research Aim

The primary aim of this research is to design, simulate, and evaluate a robust e-voting environment that synthesizes blockchain-inspired cryptographic structures for data integrity with unsupervised machine learning algorithms for the real-time detection of anomalous voting behavior.

2.2 Research Objectives

To fulfill the primary research aim, the following objectives have been established:

1. **Synthesize Scholarly Perspectives:** Conduct a critical and exhaustive literature review of 10+ scholarly papers to identify the specific failure points of current e-voting frameworks and the theoretical benefits of a hybrid Blockchain-AI approach.
2. **Conceptual Architecture Design:** Formulate a detailed system model where individual votes are treated as hashed, chained blocks, ensuring that any attempt at retroactive tampering is mathematically detectable.
3. **Develop a Simulation Environment:** Implement a Python-based simulation to generate a synthetic dataset of voting transactions, incorporating specific "fault-seeded" anomalies such as replay attacks and temporal bursts.
4. **Implement AI-Driven Anomaly Detection:** Deploy an Isolation Forest algorithm to evaluate the system's ability to isolate fraudulent behavior from legitimate human voting patterns in an unlabeled environment.
5. **Critical Comparative Analysis:** Evaluate the proposed hybrid model against global benchmarks, such as the Estonian i-Voting system, specifically addressing vulnerabilities identified in previous independent audits.

3 Literature Review

The scholarly discourse regarding e-voting is currently characterized by a tension between the promise of technological efficiency and the reality of sophisticated cyber-threats. Kshetri and Voas (2018) argues that the shift toward decentralization is not merely an upgrade but a necessity to dismantle the "centralized vulnerability profile" of modern state databases.

3.1 Decentralized Governance: The Shift Toward DLT

The fundamental appeal of Distributed Ledger Technology (DLT) in elections is its ability to distribute trust across a network rather than a single entity. Yavuz et al. (2018) demonstrates through Ethereum-based prototypes that smart contracts can automate the tallying process, ensuring the "logic" of the count is public and unalterable. This decentralization theoretically eliminates the "insider threat" where a rogue database administrator could modify records.

However, the literature is not unanimously supportive. The seminal MIT report by Specter et al. (2021) identifies the "Oracle Problem" as a significant hurdle. They argue that blockchain ensures that once a vote is on the ledger, it is immutable, but it does not

ensure that the vote was "true" at the point of origin. If a voter's device is compromised by malware, the blockchain acts as a perfect record of a fraudulent event (Park et al., 2021). This critique establishes the central thesis of this project: blockchain is a necessary "storage-layer" solution, but it must be supplemented by an "entry-layer" sentinel.

3.2 Cryptographic Integrity and the Secrecy Dilemma

Ensuring that a vote is "recorded-as-cast" requires rigorous cryptographic linking. Li and et al. (2021) demonstrate that SHA-256 hash-chaining where each block contains the digest of the previous block creates a dependency that prevents unauthorized modification. Any change to a historic block causes a "hash-mismatch" that propagates through the entire chain, alerting auditors instantly.

Yet, as Chaum (1981) originally posited, the move toward digital transparency often threatens the secrecy of the ballot. Shao and et al. (2022) notes that in public ledgers, "pseudo-anonymity" is vulnerable to metadata analysis. If a voter's wallet address or submission timestamp is linked to their real-world identity, the system fails the democratic requirement for privacy. While advanced techniques like Zero-Knowledge Proofs (ZKPs) can mitigate this, they introduce significant computational costs (Shao and et al., 2022). This research focuses on the integrity side of this trade-off, acknowledging the privacy concerns identified by Li and et al. (2021) as a critical limiting factor for real-world implementation.

3.3 AI as a Sentinel for Behavioral Anomaly Detection

Since ledgers cannot distinguish between a coerced human and a malicious script, the literature has turned to Artificial Intelligence. High-volume transaction environments make human-led audits technically unfeasible, necessitating Unsupervised Learning to detect novel fraud patterns (Agrawal and Maurya, 2022).

The "Isolation Forest" algorithm, introduced by Liu et al. (2008), is particularly effective in electoral forensics. It operates on the principle that anomalies are "few and different" and can be isolated with fewer random partitions than "normal" data points. In a voting context, bot-driven "ballot stuffing" would likely manifest as a statistical cluster multiple votes with identical timestamps or from similar network origins making them mathematically isolated from the broader human population (Chandola et al., 2009). This project synthesizes these findings by positioning AI as a "pre-ledger filter" to flag suspicious activity before it is sealed in the blockchain.

3.4 Comparative Analysis: The Estonian Benchmark and 2025 Realities

Global case studies, particularly Estonia, provide the only empirical evidence for national-scale digital voting. While Estonia is often cited as a triumph of digital governance

(Drechsler, 2020), independent audits have revealed startling vulnerabilities. Springall and et al. (2014) demonstrated that a sophisticated adversary could intercept votes at the centralized counting server, modifying the outcome without detection.

By 2025, researchers have identified over 37 potential fault scenarios in the Estonian model, ranging from client-side malware to log-file manipulation (IEEE Security Taskforce, 2025). This project’s hybrid approach directly responds to these flaws by removing the ”trusted” central server in favor of a decentralized ledger and adding AI-driven surveillance to detect the very malware attacks Springall and et al. (2014) warned about. This combination provides the ”universal verifiability” that Khader and et al. (2022) argues is the missing component in current frameworks.

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